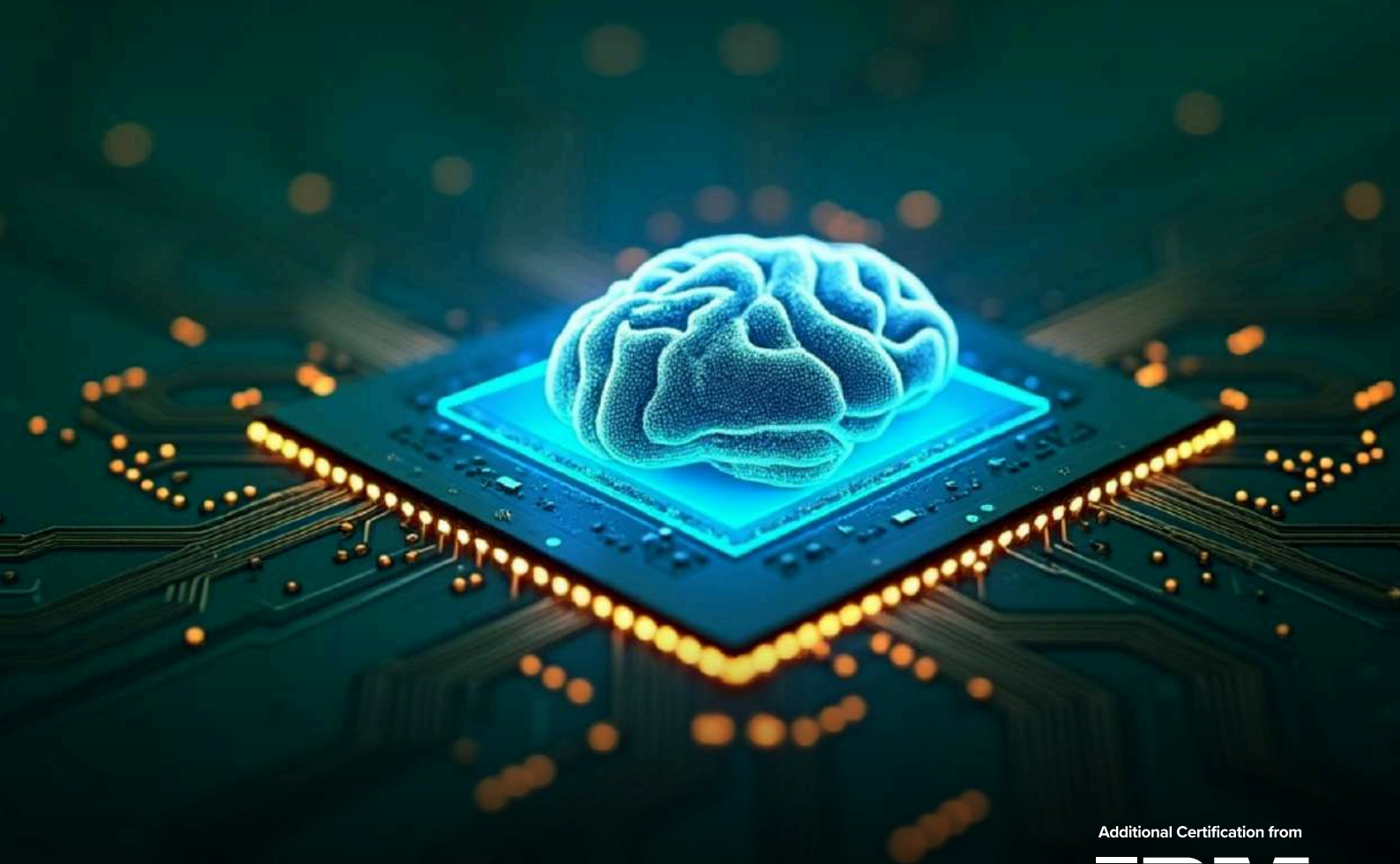


Advanced Certificate Programme in **Agentic AI and RAG Engineering**

**Design Multi-Agent AI Systems with RAG,
Memory, Tools, and Deployment**

📅 7 Months | 💻 Live Online Sessions | 🧑‍🔬 Domain Expert-Led Programme

🔧 Engineering-First Curriculum | 🏠 Optional 2-Day Campus Immersion



Additional Certification from

IBM

Programme Summary

Advanced Certificate Programme in Agentic AI and RAG Engineering

 Institute Name IITM Pravartak	 Programme Duration 7 Months	 Cost INR 1,35,000 + GST
 Pedagogy Live online weekend sessions by domain experts, along with select IITMP faculty masterclasses	 Eligibility Minimum Graduate (10+2+3); Diploma Holders with min. 5 years of work experience	 Prerequisites Programming knowledge is required
 IITM Pravartak Certification A verified digital certificate from IITM Pravartak upon successful programme completion	 3 IBM Certificates Databases with ChromaDB, Apps with LangChain and LlamaIndex, Agents with RAG and LangChain	 Faculty Weekly live sessions by domain experts, select masterclasses by IITM Pravartak lead faculty
 Weekly Effort 10-12 hours per week	 Tools Covered 15+ tools and frameworks used in modern AI and RAG engineering	 Projects 20+ advanced projects with real-world application for hands-on learning
 Capstone Project Production-oriented capstone project focused on building and deploying an enterprise-ready Agentic AI system	 Campus Immersion Optional two-day campus immersion event at IIT Madras Research Park	 Career Support Services IIMJobs pro subscription for 6 months, resume and cover letter essentials, interview prep, etc.

Learning Experience

- **Advanced RAG and Agentic AI:** 7-month engineering-first curriculum that teaches how to design, build, deploy, evaluate, and govern production-grade Agentic AI and RAG systems
- **Domain Expert-Led Course:** Designed and taught by renowned domain experts via weekly live sessions (including doubt solving)
- **Industry Agnostic Projects:** Built to equip professionals from several domain with key technical skills
- **Virtual Labs:** Integrated labs for 15+ advanced tools and libraries
- **3 IBM Certifications:** Add-on industry-credentials in the programme ecosystem to give you the professional edge
- **Flexible Learning:** Designed for working professionals for the freedom to excel in AI-dominated landscape

Frequently Asked Questions

How is the programme teaching split between IITM Pravartak Lead Faculty and domain experts?

The entire programme curriculum will be taught by Domain Experts and will also include a select live exclusive masterclasses by IITM Pravartak Lead Faculty.

What is the role of the domain experts (DEs)? Are they Institute faculty?

Domain experts will conduct live sessions where they teach each concept in depth with hands-on tools, real-world applications, and examples. They are not institute faculty but bring rich industry experience to the classroom.

Who grades/gives inputs on the assignments and projects?

The grading frameworks for assignments are developed in partnership with domain experts and the Emeritus grading team.

Is there a qualifying mark/grade and minimum attendance criteria to get the final certification in this programme?

Yes, overall 50% attendance in both domain-expert led live sessions and live faculty masterclasses are required to achieve programme completion. The qualifying mark for programme completion is 70%. Participants must also complete the capstone project in order to get the programme certification.

What if I miss the assignments for a particular week? Can I attempt them later?

If you miss an assignment for a particular week, you can complete it any time before the programme concludes. We provide flexibility for you to catch up and submit assignments at your convenience within the programme's duration.

Who are the faculty for the live masterclasses and doubt-clearing sessions?

Select live masterclasses are conducted by IITM Pravartak Lead Faculty. Weekly live sessions, including doubt-clearing sessions, are carried out by the domain experts, as they monitor individual student progress.

What if I don't find the programme appropriate for me after starting the sessions? Can I seek a refund?

We encourage participants to complete the programme to fully understand the concepts and derive valuable learning outcomes. Should you still feel the need to stop your learning journey, a refund request can be initiated before the programme commences. However, after the programme commences, the fee becomes non-refundable.

For how long will I have access to the learning materials?

You will have access to the online learning platform and all the videos and programme materials for 12 months following the programme end date. Access to the learning platform is restricted to registered participants per the terms of the agreement.

Note:

- This programme summary is provided only for your convenience. You are advised to refer to the programme brochure for more information.
- Domain expert is the programme leader responsible for conducting weekly live sessions.
- Schedule for faculty masterclass will be shared post programme orientation.

Learn Enterprise-Grade Agentic AI and RAG Engineering Like Never Before

Modern AI systems are evolving beyond simple chatbots and automation tools. The most powerful AI applications today combine two critical capabilities:



Agentic AI

autonomous AI agents that can plan, reason, interact with tools, and execute complex tasks.



Retrieval-Augmented Generation (RAG)

systems that allow AI models to retrieve and use organisational knowledge to generate accurate and context-aware responses.

While many AI programmes introduce agentic AI or RAG concepts, **very few programmes provide in-depth engineering expertise in both Agentic AI and RAG systems together.**

The **Advanced Certificate Programme in Agentic AI and RAG Engineering by IITM Pravartak** is uniquely designed to bridge this gap.

Participants will gain deep hands-on expertise in:



Designing autonomous AI agents that can plan and execute complex workflows



Building robust RAG pipelines that integrate enterprise knowledge sources



Orchestrating multi-agent systems that collaborate to solve problems



Deploying AI agents and RAG systems in real-world production environments

By combining these two advanced domains, the programme prepares professionals to build multi-agent **enterprise-grade AI architecture that is capable of autonomous reasoning and knowledge-driven decision-making.**

Growing Need for Agentic AI and RAG Skills

80% of businesses are exploring the potential of **Agentic AI**, as organisations look to automate complex workflows and decision-making processes. (Source: Deloitte's State of GenAI Report 2025)

20x Global RAG market to surpass **\$40 billion by 2035** as enterprises accelerate AI integration, growing 20.5x at a CAGR of 35.31% during the forecast period from \$1.95 billion in 2025. (Source: Researchandmarkets)

50% Professionals with **RAG engineering skills** are seeing salary increases of up to **50%**, reflecting the growing demand for engineers who can build knowledge-aware AI systems that integrate large language models with enterprise data. (Source: Glassdoor, AmbitionBox, and LinkedIn salary insights (2024-2025)).

Unlock High-Paying Roles with Enterprise-Grade AI Engineering Skills

Agentic AI Engineer

+35-55% Higher Pay

RAG Engineer

+40-60% Higher Pay

LLM Systems Engineer

+40-65% Higher Pay

AI Platform Engineer

+30-50% Higher Pay

AI Architect

+45-70% Higher Pay

- Note:
- Source of salary data: Glassdoor, AmbitionBox, and LinkedIn salary insights (2024-2025).
 - These salary figures are for a similar work experience bracket.
 - Salaries for AI roles in India are among the highest in the country. Your salary will vary based on your skills, experience, and the city where you work.

Who Should Enrol in This Programme?

This programme is designed for professionals across engineering, data, and technology-driven roles who are ready to architect, build, and deploy production-grade Agentic AI and RAG systems in real-world environments.

- **Data Science, AI, and Engineering Professionals**

Software engineers, AI/ML engineers, data engineers, backend developers, and solution architects looking to move beyond model experimentation into designing multi-agent systems, advanced RAG pipelines, and deployable AI services using Python and modern orchestration frameworks.

- **Product and Technical Leaders**

Technical product managers, platform leads, and innovation heads responsible for integrating AI into enterprise systems and seeking structured expertise in agent architectures, evaluation, deployment, and governance.

- **Mid-Career Technology Professionals**

Professionals with prior exposure to programming, APIs, or backend systems who want to upskill into Agentic AI engineering and transition into high-impact AI system design roles.

By the end of the programme, participants will gain practical expertise in designing, building, and deploying AI systems that can reason, plan, and act autonomously.



Advance Your Career with Agentic and RAG Skills

Organisations are hiring for skills in enterprise multi-agent architecture i.e. building smarter AI systems for real-world enterprises. This programme combines deep technical learning with hands-on projects, modern tools, and industry certifications to build production-ready AI agents and knowledge-driven AI systems.



Most Comprehensive Agentic AI and RAG Engineering Curriculum*

Gain in-depth expertise in two of the most critical areas of modern AI - designing autonomous agents and building RAG systems that integrate enterprise knowledge with large language models.



Learn from Experienced Domain Experts and IITM Pravartak Lead Faculty

Receive mentorship through weekly live sessions led by domain experts and a select live IITM Pravartak Lead Faculty masterclasses - offering practical, real-time engagement.



Engineer Production-Ready AI Systems

Design, deploy, and monitor enterprise-grade AI agents and RAG pipelines using modern frameworks, orchestration patterns, and production engineering practices.



Work on 20+ Hands-On Assignments and Projects

Apply key concepts through practical projects that simulate real-world AI engineering workflows and enterprise use cases.



Develop a Production-Oriented Capstone Project

Design and deploy a complete AI agent system that integrates multi-agent orchestration, knowledge retrieval, and deployment capabilities.



Gain Hands-On Experience with 15+ Modern AI Tools

Work with industry-relevant frameworks, vector databases, and deployment platforms used to build real-world AI applications.

**Note:*

- The entire programme curriculum will be taught by Domain Experts and will also include a select live exclusive masterclasses by IITM Pravartak Lead Faculty.



Earn an Industry-Recognised Certificate from IITM Pravartak

Receive a certificate of completion from IITM Pravartak upon successfully completing the programme requirements.



Elevate Your Resume with Three Additional IBM Certifications

Strengthen your professional credentials with three IBM certifications in RAG, LangChain and AI Agent Development



Experience Optional Campus Immersion at IIT Madras Research Park

Participate in an optional two-day campus immersion event and experience the innovation ecosystem at IIT Madras Research Park.

Note:

- The entire programme curriculum will be taught by Domain Experts and will also include a select live exclusive masterclasses by IITM Pravartak Lead Faculty.

System Requirements to Pursue the Programme

- Laptop/desktop with Windows 10/11 (64-bit) or Ubuntu 20.04+ or macOS 12 or above, minimum 8 GB RAM (16 GB recommended)
- 4-core Intel i5/i7 or AMD equivalent or Apple Silicon (M1/M2/M3 or above),
- 120 GB free SSD storage for windows/linux or 30GB free disk space for MacOS
- Ability to install Python, Docker, and VM
- Stable internet connection

Note:

- Learners must have a personal system meeting minimum hardware and virtualisation requirements to run hands-on labs. Company-issued, restricted, or low-spec devices (e.g., 8 GB RAM) may not support all lab activities.

The Edge You Need: How is This Agentic AI and RAG Engineering Programme Different?

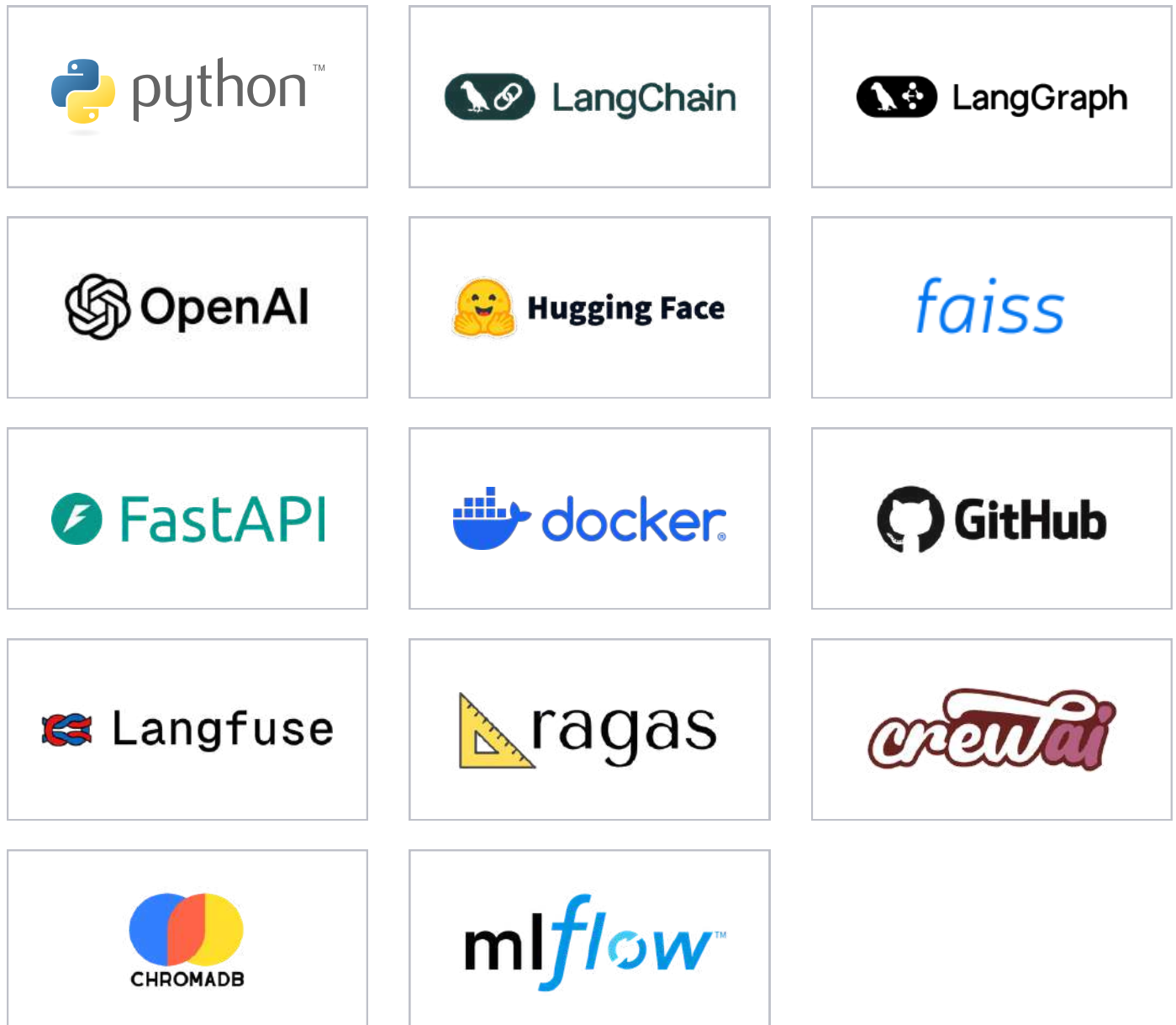
Programme Features	Advanced Certificate Programme in Agentic AI and RAG Engineering	Other Outdated/Non-Accredited GenAI and Agentic AI Programmes
Certification from a top ranked institution	Industry certificate from IITM Pravartak; 3 IBM certifications focused on RAG, LangChain, and AI Agent Engineering	Certification from non-accredited or low ranked institutes and additional certifications are rarely offered or come with add-on costs
Programme structure	Structured 7-month engineering-first curriculum designed for building and deploying production-grade Agentic AI systems	Short-duration programmes primarily focused on prompting or basic RAG concepts without full systems integration
Depth of curriculum	Comprehensive coverage across reasoning workflows, multi-agent orchestration, memory tiers, evaluation frameworks, observability, and scalable deployment stacks	Conceptual and theory-driven learning with introductory exposure to embeddings, agents, and context handling
Capstone and projects	End-to-end production capstone with hands-on assignments covering multi-agent systems, advanced RAG pipelines, and deployment workflows	1-2-week capstone and limited projects focused on isolated features without lifecycle deployment
Advanced Tools and Frameworks	Hands-on execution using industry-grade frameworks such as LangChain, LangGraph, CrewAI, OpenAI API, vector databases, FastAPI, and Docker	High-level automation or low-code tools taught under workflow-first approaches with limited production exposure

Note:

- The entire programme curriculum will be taught by Domain Experts and will also include a select live exclusive masterclasses by IITM Pravartak Lead Faculty.
- Schedule for faculty masterclass will be shared post programme orientation.

More than 15 Agentic AI and RAG Tools Covered

Participants will work with modern tools and frameworks used in AI engineering and agent development.



Note:

- Programme learning hours and tools are subject to change; updated details will be shared closer to the start date.
- The list of tools covered is not comprehensive. Final details will be shared closer to the start date.
- All product and organisation names are trademarks or registered trademarks of their respective holders, and their use does not imply any affiliation with or endorsement by them.

Industry-Recognised Certificate from IITM Pravartak

Participants will be awarded a completion certificate on successful completion of the programme.



Note:

- All certificate images are for illustrative purposes only and may be subject to change at the discretion of, IITM Pravartak.
- To receive the completion certificate, the participants must score a minimum of 70% in their exams and successfully complete the capstone project.
- Overall, 50% attendance in both domain-expert led live sessions and live faculty masterclasses are required to achieve programme completion.

IBM Certificates

Participants who successfully complete the programme will receive three IBM completion certificates as well as a certificate from IITM Pravartak. They will be certified in the following topics:



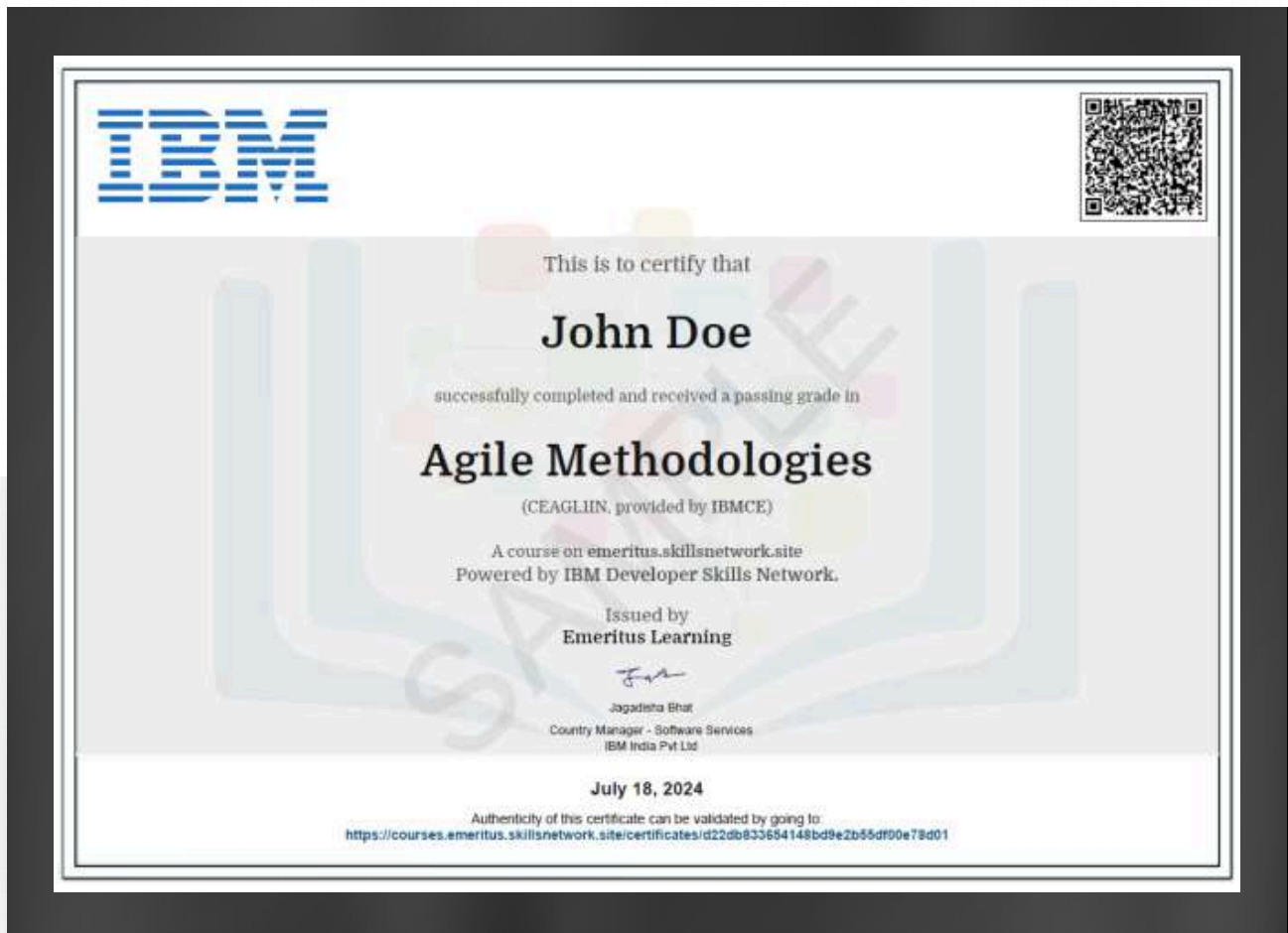
RAG: Vector
Databases with
ChromaDB



RAG: Build Apps
with LangChain and
LlamaIndex



Building AI
Agents with RAG
and LangChain



Note:

- All certificate images are for illustrative purposes only and may be subject to change at the discretion of IBM.

Programme Pillars



Pillar 1: Python Programming Refresher

- ▶ Python Programming Refresher



Pillar 2: AI Strategy

- ▶ AI Problem Framing and Agent Use-Case Design



Pillar 3: Agent Foundations

- ▶ Introduction to Agents
- ▶ Agentic Reasoning for Engineers



Pillar 4: Prompt Engineering and Agent Cognition

- ▶ Prompt Systems and Personas
- ▶ Iterative Prompt Development and Feedback Loops
- ▶ Prompt Chaining in Code



Pillar 5: Workflow Design and Orchestration

- ▶ Agentic Workflow Design
- ▶ Routing and Parallelisation
- ▶ State and Memory Fundamentals
- ▶ Reliability by Design



Pillar 6: Tools, Data, and Integration

- ▶ Tool Use and Function Calling
- ▶ Structured Outputs and Data Operations
- ▶ External Integrations and Web Search
- ▶ Databases and CRUD Agents



Pillar 7: Retrieval-Augmented Agents

- ▶ RAG Foundations and Agentic RAG
- ▶ Multi-RAG Patterns
- ▶ Memory Tiers and Long-Term Memory
- ▶ Evaluation for RAG and Agents



Pillar 8: Multi-Agent Intelligence

- ▶ Multi-Agent Systems
- ▶ Multi-Agent State Coordination
- ▶ Multi-Agent Orchestration
- ▶ Multi-Agent RAG and Tool Mesh



Pillar 9: LLM Ops, Deployment, Observability and Governance (LLM Ops)

- ▶ Productionisation: Services and MCP
- ▶ Observability and Cost Control
- ▶ Security, Privacy, and Governance
- ▶ CI/CD, Testing and Launch

Note:

- All programme curriculum stated here is subject to change as per the discretion of IITM Pravartak, Emeritus, or IBM.



Programme Modules



Module 1: Python Programming Refresher

- ▶ Python execution model
- ▶ Environment and package management
- ▶ Core language constructs and file handling
- ▶ API interaction and asynchronous programming
- ▶ FastAPI services
- ▶ Embeddings intuition
- ▶ Testing with pytest
- ▶ LLM-assisted coding workflows
- ▶ Lightweight deployment with monitoring



Module 2: Applied Python and AI System Thinking

- ▶ Prototyping AI and agent workflows in Python
- ▶ AI transformation overview
- ▶ Automation vs agentic systems
- ▶ Identifying agent-worthy problems
- ▶ ROI and feasibility analysis
- ▶ Human-in-the-loop design and risk classification
- ▶ Enterprise adoption patterns
- ▶ Translating business problems into agent requirements



Module 3: Introduction to Agents

- ▶ What are AI agents
- ▶ Agentic AI vs traditional systems
- ▶ Agent lifecycle and capabilities
- ▶ Levels of autonomy in AI systems
- ▶ Real-world examples of agents



Module 4: Agentic Reasoning for Engineers

- ▶ Agent vs application mental models
- ▶ Task decomposition
- ▶ Reasoning strategies (CoT, ToT, ReAct)
- ▶ Prompt structure design
- ▶ Structured outputs with Pydantic
- ▶ Reliability-first mindset



Module 5: Prompt Systems and Personas

- ▶ Role and persona prompting
- ▶ Instruction hierarchies

- ▶ Safety boundaries
- ▶ Deterministic outputs using schemas
- ▶ Few-shot prompt libraries



Module 6: Iterative Prompt Dev and Feedback Loops

- ▶ Critic-creator loops
- ▶ Self-refine and verification patterns
- ▶ Unit-test-driven prompting
- ▶ Automated critique rubrics
- ▶ Failure-mode catalogues



Module 7: Prompt Chaining in Code

- ▶ Prompt chaining patterns
- ▶ Guardrails and validators
- ▶ Schema validation with Pydantic
- ▶ Retries and error handling
- ▶ Minimal orchestration in Python



Module 8: Agentic Workflow Design

- ▶ Orchestrator-worker architecture
- ▶ Evaluator-optimiser loops
- ▶ Router patterns
- ▶ Sequential vs parallel vs conditional workflows
- ▶ Workflow design diagrams



Module 9: Routing and Parallelisation

- ▶ Classifiers and intent routers
- ▶ Routing strategies
- ▶ Parallel fan-out/fan-in pipelines
- ▶ Aggregation and conflict resolution
- ▶ Idempotent workflows



Module 10: State and Memory Fundamentals

- ▶ Short-term state vs working memory
- ▶ Message and state graphs
- ▶ Ephemeral vs persistent memory
- ▶ Context window management
- ▶ Token budgeting



Module 11: Reliability by Design

- Specification-driven agents
- Testing strategies
- Invariants and safety checks
- Exception-handling strategies
- SLAs and SLOs for agent systems



Module 12: Tool Use and Function Calling

- Tool schemas
- Secure tool adapters
- Tool selection strategies
- Loop prevention
- API rate limits
- Retries and timeouts



Module 13: Structured Outputs and Data Ops

- Pydantic models and structured outputs
- Function-calling patterns
- JSON/Avro data pipelines
- Schema validation and versioning



Module 14: External Integrations and Web Search

- REST and GraphQL integrations
- Authentication flows and secrets management
- Web search agents
- Grounding and citation strategies



Module 15: Databases and CRUD Agents

- Text-to-SQL agents
- Database interaction patterns
- Constrained updates and transaction safety
- Audit logging
- Vector database fundamentals



Module 16: RAG Foundations and Agentic RAG

- Document indexing and chunking
- Embeddings and vector search
- Hybrid retrieval strategies
- Query reformulation

- ▶ Reranking
- ▶ Hallucination control



Module 17: Multi-RAG Patterns

- ▶ Multi-corpus retrieval systems
- ▶ Multi-tool retrieval orchestration
- ▶ Caching strategies
- ▶ Freshness and knowledge drift management



Module 18: Memory Tiers and Long-Term Memory

- ▶ Semantic, episodic, and procedural memory
- ▶ Embeddings hygiene
- ▶ Session stitching
- ▶ Privacy and lifecycle policies
- ▶ Summarisation pipelines



Module 19: Evaluation for RAG and Agents

- ▶ Golden datasets and synthetic evaluation
- ▶ RAGAS and evaluation frameworks
- ▶ Step-wise vs outcome evaluation
- ▶ Cost-quality trade-offs



Module 20: Multi-Agent Systems

- ▶ Multi-agent architectures
- ▶ Agent roles and capabilities
- ▶ Shared tools
- ▶ Communication patterns
- ▶ Resource contention control



Module 21: Multi-Agent State Coordination

- ▶ Global vs local state management
- ▶ Locks and leases
- ▶ Conflict detection
- ▶ Consensus strategies
- ▶ Concurrent agent coordination



Module 22: Multi-Agent Orchestration

- ▶ Planner-executor models
- ▶ Subgoal generation

- Supervisor agents
- Human-in-the-loop escalation
- Failure-recovery strategies



Module 23: Multi-Agent RAG and Tool Mesh

- Specialised retrievers
- Synthesis agents
- Cross-agent memory sharing
- Multi-agent knowledge systems
- Load testing and red-teaming



Module 24: Productionisation: Services and MCP

- MCP servers and tools
- Secure tool exposure
- FastAPI services
- Containerisation
- Infrastructure choices (serverless vs Kubernetes)



Module 25: Observability and Cost Control

- System tracing
- Metrics and logging
- Prompt and version tracking
- Token usage monitoring
- Rate-limit strategies



Module 26: Security, Privacy, Governance

- Data governance and PII protection
- Policy enforcement
- Prompt injection defence
- Tool misuse prevention
- Red-team playbooks



Module 27: CI/CD, Testing and Launch

- Unit and integration testing for agents
- Offline and online evaluation gates
- CI/CD pipelines
- Blue-green deployment
- Rollback strategies
- Production monitoring

Note:

- The entire programme curriculum will be taught by Domain Experts and will also include a select live exclusive masterclasses by IITM Pravartak Lead Faculty.
- All programme curriculum - topics, modules, submodules, tools — stated here are subject to change as per the discretion of IITM Pravartak or Emeritus

IBM Certificate Modules



Certificate 1: RAG - Build Apps with LangChain and LlamaIndex

- ▶ Introduction to RAG
- ▶ Build Applications with RAG
- ▶ Build RAG Applications with LlamaIndex



Certificate 2: RAG - Vector Databases with ChromaDB

- ▶ Introduction to Vector Databases and Chroma DB
- ▶ Vector Databases for Recommendation Systems and RAG



Certificate 3: Building AI Agents with RAG and LangChain

- ▶ RAG Framework
- ▶ Prompt Engineering and LangChain

Note:

- All programme curriculum stated here is subject to change as per the discretion of IITM Pravartak, Emeritus, or IBM.

Projects and Assignments



Python Programming Refresher

Build a production-ready Python AI project including APIs, embeddings utilities, FastAPI services, testing pipelines, and LLM-assisted coding workflows.



AI Problem Framing and Agent Use-Case Design

Identify a high-value agent use case, design an agent solution canvas, and define risk controls, autonomy levels, and success metrics.



Introduction to Agents

Analyse real-world agent systems and map the lifecycle of an AI agent from input to reasoning, tool use, memory, and feedback loops.



Agentic Reasoning for Engineers

Design structured reasoning workflows using CoT/ReAct patterns and build a reliable reasoning pipeline with structured outputs and validation.



Prompt Systems and Personas

Create persona-based prompt systems with structured schemas and reusable few-shot example libraries.



Iterative Prompt Dev and Feedback Loops

Implement prompt refinement workflows using critic-creator loops and automated evaluation rubrics.



Prompt Chaining in Code

Build prompt chaining pipelines with guardrails, schema validation, retries, and controlled execution flows.



Agentic Workflow Design

Design and compare agent workflow architectures such as orchestrator-worker, router, and evaluator-optimizer patterns.



Routing and Parallelization

Implement routing systems and parallel execution workflows with aggregation and conflict resolution strategies.



State and Memory Fundamentals

Design agent state management and implement a lightweight memory system with context management and summarization.



Reliability by Design

Define reliability specifications, failure taxonomies, and performance SLAs for production-grade agent systems.



Tool Use and Function Calling

Build secure tool integrations and implement function-calling agents with policies for tool selection and execution control.



Structured Outputs and Data Ops

Develop structured-output pipelines with schema validation, versioning, and downstream data integration.



External Integrations and Web Search

Design and integrate external APIs and web search tools with grounding and citation strategies.



Databases and CRUD Agents

Build database-aware agents capable of safe text-to-SQL operations and transaction-controlled updates.



RAG Foundations and Agentic RAG

Develop an agentic RAG pipeline with hybrid retrieval, reranking, query reformulation, and hallucination control.



Multi-RAG Patterns

Design multi-source retrieval systems with routing, caching, and knowledge freshness management.



Memory Tiers and Long-Term Memory

Implement long-term memory architectures with semantic, episodic, and summarization-based memory systems.



Evaluation for RAG and Agents

Build evaluation datasets and measure agent performance using structured metrics and cost-quality analysis.



Multi-Agent Systems I

Design and implement a collaborative multi-agent system with defined roles, shared tools, and communication protocols.



Multi-Agent State Coordination

Implement coordination mechanisms for concurrent agents using locks, leases, and conflict resolution strategies.



Multi-Agent Orchestration

Build planner-executor workflows with supervisor agents and human-in-the-loop escalation mechanisms.



Multi-Agent RAG and Tool Mesh

Develop a multi-agent tool mesh where agents retrieve, synthesize, and share knowledge across systems.



Productionisation: Services and MCP

Package agent systems as deployable services with APIs, containerization, and tool interfaces.



Observability and Cost Control

Implement tracing, monitoring, and cost tracking for agent workflows with token and API usage visibility.



Security, Privacy, Governance

Design governance policies and defenses against prompt injection, tool misuse, and data risks.



CI/CD, Testing and Launch

Create testing pipelines, evaluation gates, and deployment workflows for safely launching AI systems.

Note:

- All programme curriculum stated here is subject to change as per the discretion of IITM Pravartak, Emeritus, or IBM.

IITM Pravartak Lead Faculty Profile



Prof. Madhusudhanan Baskaran
Programme Director

Prof. Madhusudhanan is a Principal AIML Consultant and faculty at IITM Pravartak, specialising in Agentic AI, generative systems, and intelligent automation. With over 32 years of experience across academia, industry, and government, he brings deep expertise in designing AI agents that plan, reason, and adapt autonomously.

His portfolio includes AI-led projects for the Supreme Court of India, CAG, ReBIT, and the Indian Army, focusing on responsible, scalable AI systems using LLMs, speech technologies, and document intelligence. A key voice in India's AI transformation, he actively mentors deep-tech startups and leads initiatives in explainable AI and modular agent architectures.

Publications:



The Scientific World Journal: "Synchronous firefly algorithm for cluster head selection in WSN"



International Conference on Computer and Information Sciences: "Improving security in Wireless Sensor Network using trust and metaheuristic algorithms"

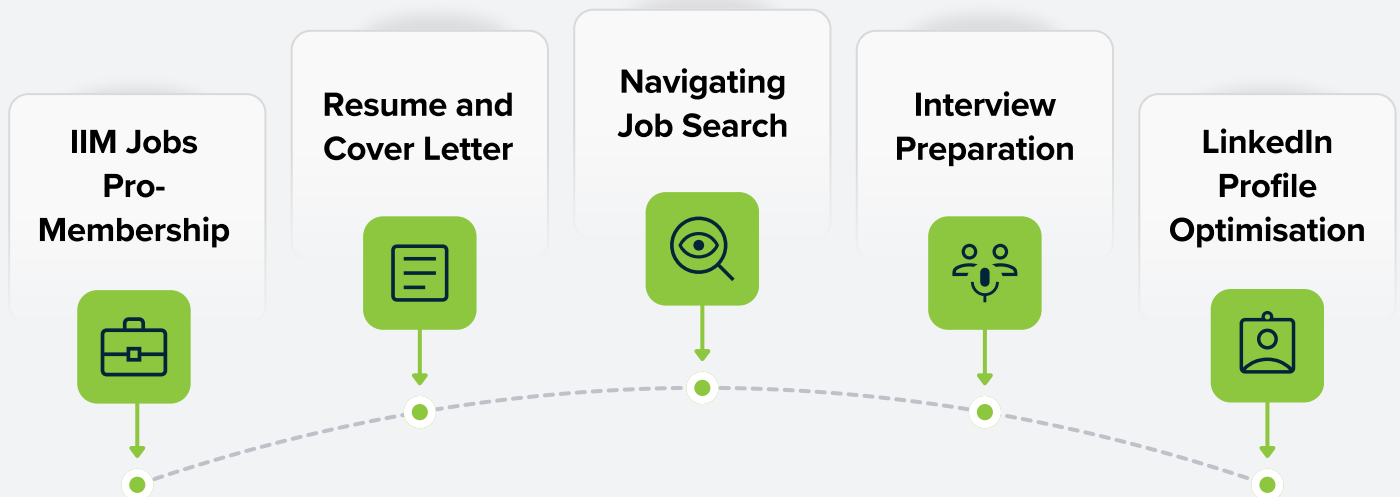


Soft Computing 24: "Computerised grading of brain tumors supplemented by artificial intelligence"

Emeritus Career Services Benefits

Fifteen recorded sessions and resources in the following categories

(Please note: These sessions are not live):



Key benefits:

- **Pro-Membership and features of IIMJobs:** Access to job insights recruiter action status, follow-up actions, and ability to chat with recruiters who shortlist your profile. Membership will be valid for 6 months and provided at the end of the programme
- **Spotlight on IIMJobs and Hirst:** Profile boost for applied jobs (that align with acquired certification), greater profile visibility - highlighted with institute name along with a testimony of certificate acquisition by the candidate
- **Spotlight Plus:** All the benefits of Spotlight and added advantages like profile and rank boost in the recruiter search database
- **Resume builder tool:** 6-month access to DIY resume builder, auto resume creator, optimization suggestions based on key parameters, guide on information to be incorporated, and unlimited resume iterations within the duration.

Notes:

- IITM Pravartak or Emeritus do NOT promise or guarantee a job or progression in your current job. Career Services are only offered as a service that empowers you to manage your career proactively.
- The Career Services mentioned here are offered by Emeritus. IITM Pravartak is NOT involved in any way and makes no commitments regarding the Career Services mentioned here.
- This service is available only for Indian residents enrolled into selected Emeritus programmes.

Programme Details



Programme Duration

7 Months



Programme Start Date

29th June 2026



Programme Fee

INR 1,35,000 + GST



Payment Options

Basic instalment plans



Special Pricing

Up to 10% fee benefit for corporate plans



Programme Format

Curriculum and weekly live sessions by domain experts, select live faculty masterclasses, IIT Madras Research Park campus immersion for 2 days (optional)*

**Note:*

- Only participants who have successfully completed the programme will be allowed to visit the campus.

Eligibility Criteria:

Minimum Graduate; Diploma Holders with min. 5 years of work experience; Programming knowledge required

**Note:*

- Only participants who have successfully completed the programme will be allowed to visit the IITM Research Park campus.
- The immersion will only be conducted with a minimum number of learners signing up.
- Schedule for faculty masterclass will be shared post programme orientation.
- The entire programme curriculum will be taught by Domain Experts and will also include a select live exclusive masterclasses by IITM Pravartak Lead Faculty.

About IITM Pravartak

IITM Pravartak is funded by the Department of Science and Technology, Government of India, under its National Mission on Interdisciplinary Cyber-Physical Systems, and hosted as a Technology Innovation Hub (TIH) by IIT Madras. The NM-ICPS is a comprehensive Mission aimed at complete convergence with all stakeholders by establishing strong linkages between academia, industry, Government, and International Organisations.

About Emeritus

Emeritus is committed to teaching the skills of the future by making high-quality education accessible and affordable to individuals, organisations and governments worldwide. It does this by collaborating with more than 80 top-tier universities across the United States, Europe, Latin America, Southeast Asia, India, and China. Emeritus's short courses, degree programmes, professional certificates, and senior executive programmes help individuals learn new skills and transform their lives, companies, and organizations. Its unique model of state-of-the-art technology, curriculum innovation; and hands-on instruction from senior faculty, mentors, and coaches has educated more than 350,000 individuals across more than 80 countries.



Apply for the programme here

APPLY NOW

Refer your colleague and earn benefits

REFER NOW

Schedule a call with a programme advisor

KNOW MORE



WhatsApp an Advisor on +918657038243*

**This number does not accept any calls. Please message your queries.*

For registration and any other information,
please get in touch with us at tech.certificates@emeritus.org

Erulearning Solutions Private Limited (a company incorporated in India) is a subsidiary of Eruditus Learning Solutions Pte Ltd (a company incorporated in Singapore), and operates under the brand name of 'Eruditus' and 'Emeritus'.

In collaboration with

